Several years later, Einstein came back with yet another thought experiment to demonstrate his concerns about locality, one that would resound for decades. Einstein and two colleagues, Boris Podolsky and Nathan Rosen, published a paper in 1935 with the provocative title “Can Quantum Mechanical Description of Physical Reality Be Considered Complete?” This paper, known as the EPR paper after its authors’ initials, is sometimes depicted as Einstein’s last desperate move in his showdown with Bohr. The truth is far messier—and much more intriguing.

The EPR paper, on the face of it, isn’t about locality—ironically, it appears to be a way around the Heisenberg uncertainty principle. But instead of devising a way to measure the momentum and position of a single particle directly at a single time, as Einstein had purportedly done in his earlier thought experiments, the EPR paper goes about it indirectly. The thought experiment at the heart of the paper imagines a pair of particles, A and B, which collide head-on, interact in a very specific and delicate way, then fly off in two opposite directions. Momentum is always conserved—it’s a basic law of nature—so the total momentum of the particles is fixed over time. And, because of the way the particles interact, the distance between them at any time is easy to calculate.

In Newton’s physics, this would be like two identical billiard balls hitting each other head-on, then bouncing off to opposite ends of an enormous pool table. Because the total momentum of the two balls has to be zero, knowing one ball’s speed and direction would instantly tell you the other ball is moving at the same speed in the opposite direction. Similarly, finding one ball would allow you to calculate the location of the other ball, if you know the time and location of the collision.

In quantum physics, the situation is a little trickier. According to the Copenhagen interpretation, particles don’t have properties like position or momentum (or anything else) until those properties are measured. But, EPR argued, measurements made on one particle couldn’t instantly affect another particles far away. So, to get around the uncertainty principle, just wait until particles A and B are very far apart, then find the momentum of A. Measuring A’s momentum lets you infer B’s momentum without disturbing B at all. Then simply measure the position of B. Now you know B’s position and momentum, to arbitrary precision, at the same time. Therefore, argued EPR, a particle can have a definite position and